



Subject Name: Programming for Problem Solving
Subject Code: BE01R00121

Faculties: Prof. Rashid Khan, Prof. Ashasri Parmar, Prof. Vaibhavi Joshi

Sr. No	CHAPTER NO- 1: INTRODUCTION TO COMPUTER AND PROGRAMMING	Marks
	TOPIC:1 Introduction, Basic Block Diagram and Functions of Various Components of Computer	
	SHORT QUESTIONS	
1	Give full form of RAM and ROM. [NLJIET]	1
	Read-only memory, Random access memory	
2	Define: Compiler.(Aug-2022-NEW)[NLJIET]	1
	The compiler is software that converts a program written in a high-level language (Source Language) to a low-level language (Object/Target/Machine Language/0's, 1's).	
3	What do you mean by program and program control?(Aug-2022-NEW)[NLJIET]	1
	A program is a set of instructions that a computer uses to perform a specific function. Program control is how a program makes decisions or organizes its activities. Program control typically involves executing particular code based on the outcome of a prior operation or a user input.	
4	Programming needs logic building. Justify. (Aug-2022-NEW)[NLJIET]	1
	Programming needs logic building because a program is a set of instructions that solve a problem. To write effective and efficient code, a programmer must understand the problem they are trying to solve and use a logical and organized approach to write the code. A well-built logic in programming helps in debugging the code, reducing the chance of errors, and making the code more maintainable as well as effective	
5	Define Interpreter.(Aug-2022-NEW)[NLJIET]	1
	An interpreter is a computer program, which converts each high-level program statement into the machine code. The conversion of the high level instructions is performed line-by-line to machine level language.	
	DESCRIPTIVE QUESTIONS	
1	Draw and explain the block diagram of Computer System. (Dec-2013-OLD) [NLJIET]	7
	Name the major components of computer system and give their function. (May-2018-OLD) [NLJIET]	3
	Categorize the major components of computer system and give their function. (Jan-2020-NEW) [NLJIET]	4
	Draw block diagram of computer system and explain the functions of each component in detail.(Aug-2022-NEW)[NLJIET]	7
	List out and briefly explain basic building blocks of Computer system. (Mar-2023-NEW)[NLJIET]	3
	Design block diagram of computer and explain functionalities of various components of	3

computer.(Aug-2023-NEW)[NLJIET]

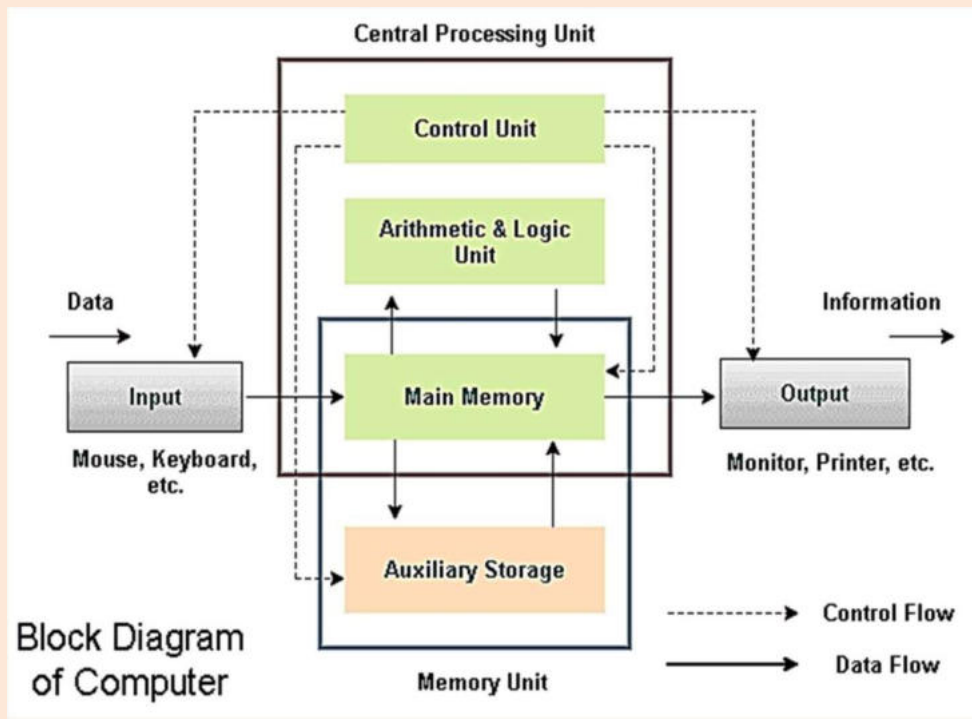
Briefly explain different components of Computer system(Jan-2024-NEW)[NLJIET]

Explain basic block diagram and functions of various components of computer.

(Aug-2023-OLD)[NLJIET]

7

3



- **Central Processing Unit (CPU)**
- The CPU can be called the brain of a computer system.
- The central processing unit (CPU) is where the majority of the important calculations and comparisons are performed. In addition, the CPU is in charge of turning on and controlling the operation of the other units.
- The arithmetic logic unit, also known as the ALU, and the control unit (CU) are the two primary elements that make up this unit. First, let's take a moment to briefly define these two different units of CPU.
- **Arithmetic Logic Unit (ALU)**
- The arithmetic logic unit is responsible for carrying out all of the mathematical operations, including addition, subtraction, multiplication, and division. In addition to that, a logical operation is used for the comparison.
- **Control Unit (CU)**



- In addition, the control unit of a central processing unit is responsible for directing the overall operation of a computer. In addition to this, it exercises control over all devices connected to the CPU, including memory and input/output devices.
- The CU is responsible for retrieving instructions from memory, decoding those instructions, interpreting those instructions to determine what tasks are to be carried out, and then sending appropriate control signals to the other components so that they can carry out the steps necessary to execute the instruction.
- **Input/Output Unit**
 - The input/output unit is made up of different devices that are responsible for transmitting and receiving information between the memory of the computer and the outside world.
 - The information that is entered into the computer via the input unit is saved in the memory of the device for later processing. The completed processing can then be saved in the memory and either recorded or displayed on the output medium.
- **Memory Unit**
 - Memory units are an integral part of any modern digital computer. It is the repository for all of the results, both intermediate and final.
 - The data that are read from the primary storage or an input unit are moved to the memory of the computer so that they can be processed. These data can come from either the main storage or an input unit.
 - The data that needs to be processed and the instructions that need to be carried out are both stored in this memory unit so they can be accessed quickly.
 - This storage unit is divided into two more sub units:
 - Primary memory
 - Secondary memory.
 - **Primary memory:**
 - This is a type of memory storage where the most recent memory is stored. In this memory only main and running activities are stored, it is smaller in size than secondary memory.
 - Data in this unit is stored temporarily and the data gets wiped out once the power is cut off, so this is also called as temporary memory or main memory.
 - Many of have already heard the memory type called “RAM”. So, this is also an example of primary memory.
 - This memory can be directly accessed by the CPU whenever needed for the processes.



	• This memory is also used for reading and writing purposes. • If any operation is to be done on the data by the CPU, then initially that data is stored in the primary memory i.e., RAM and then processed. • <u>Secondary Memory:</u> • As the data stored in the primary memory is temporary, thus to stored permanent data secondary data is used. • For permanent data storage secondary memory is used. This provides the primary memory data for processing. Secondary memory is also called as auxiliary memory. • Hard disks are the examples of the secondary memory. This memory type is less expensive than the primary memory.																	
	TOPIC:2 Concepts of Hardware and Software, Types of Software, Compiler and Interpreter																	
	SHORT QUESTIONS																	
1	Give difference between hardware and software. [NLJIET]	1																
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1	Define following: 1) Operating System 2) Compiler 3) Interpreter 4) Assembler 5) Loader 6) Linker 7) Pre-processor. (Jun-2014-OLD) [NLJIET]	7																
	Recall the definitions of (I) Assembler (II) Compiler (III) Interpreter. (Jan-2017-OLD) [NLJIET]	3																
	Write the function of (I) Assembler (II) Compiler (III) Interpreter. (Jan-2019-OLD) [NLJIET]	3																



	<u>Operating Systems</u> lies in the category of system software. It basically manages all the resources of the computer. An operating system acts as an interface between the software and different parts of the computer or the computer hardware. The operating system is designed in such a way that it can manage the overall resources and operations of the computer. Examples of operating systems are Windows, Linux, Mac OS, etc. An operating system (OS) is a collection of software that manages computer hardware resources and provides common services for computer programs. <u>The compiler</u> is software that converts a program written in a high-level language (Source Language) to a low-level language (Object/Target/Machine Language/0, 1's). <u>An interpreter</u> is a computer program, which converts each high-level program statement into the machine code. This includes source code, pre-compiled code, and scripts. <u>Assembler</u> is a program for converting instructions written in low-level assembly code into relocatable machine code and generating along information for the loader. <u>A linker</u> is an important utility program that takes the object files, produced by the assembler and compiler, and other code to join them into a single executable file. In computer systems <u>a loader</u> is the part of an operating system that is responsible for loading programs and libraries. It is one of the essential stages in the process of starting a program, as it places programs into memory and prepares them for execution. <u>The C preprocessor</u> is a macro preprocessor (allows you to define macros) that transforms your program before it is compiled. These transformations can be the inclusion of header files, macro expansions, etc. <u>All preprocessing directives</u> begin with a # symbol. For example, #define PI 3.14 The #include preprocessor is used to include header files to C programs. For example, #include <stdio.h>	
2	What is Software and Hardware? Explain different types of Software. (Jun-2015-OLD) (Dec-2015-OLD) [NLJIET] List out types of software with examples. (Jun-2016-OLD) (Jan-2020-NEW) [NLJIET]	3 3

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Hardware refers to the physical components of a computer. Computer Hardware is any part of the computer that we can touch these parts. These are the primary electronic devices used to build up the computer. Examples of hardware in a computer are the Processor, Memory Devices, Monitor, Printer, Keyboard, Mouse, and Central Processing Unit.

Types of Computer Hardware

- Input Devices
- Output Devices
- Storage Devices
- Internal Component

Input Devices: Input Devices are those devices through which a user enters data and information into the Computer or simply, User interacts with the Computer. Examples of Input Devices are Keyboard, Mouse, Scanner, etc.

Output Devices: Output Devices are devices that are used to show the result of the task performed by the user. Examples of Output Devices are Monitors, Printers, Speakers, etc.

Storage Devices: Storage Devices are devices that are used for storing data and they are also known as Secondary Storage Data. Examples of Storage Devices are CDs, DVDs, Hard Disk, etc

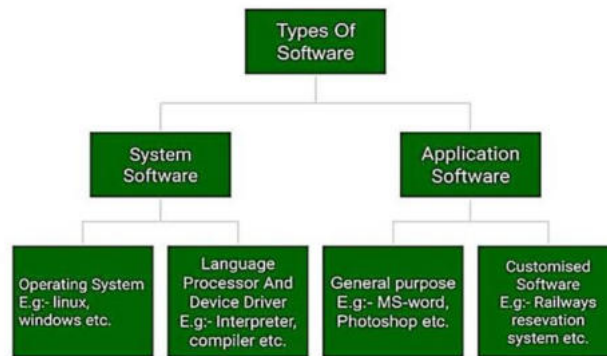
Internal Component: Internal Components consists of important hardware devices present in the System. Examples of Internal Components are the CPU, Motherboard, etc.

Software is a collection of instructions, procedures, and documentation that performs different tasks on a computer system. we can say also Computer Software is a programming code executed on a computer processor. The code can be machine-level code or code written for an operating system. Examples of software are MS- Word, Excel, PowerPoint, Google Chrome, Photoshop, MySQL, etc.

Types of Computer Software

- System Software
- Application Software

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System Software	Application Software
System software are mainly designed for managing system resources.	Application software are designed to accomplish tasks for specific purposes.
Programming of system software is complex.	Programming of application software is comparatively easy.
A computer cannot run without system software.	A computer can easily run without application software.
System software do not depend on application software.	Application software depend on system software and cannot run without system software.

System Software: System Software is a component of Computer Software that directly operates with Computer Hardware which has the work to control the Computer's Internal Functioning and also takes responsibility for controlling Hardware Devices such as Printers, Storage Devices, etc. Types of System Software include Operating systems, Language processors, and Device Drivers.

Application Software: Application Software are the software that works the basic operations of the computer. It performs a specific task for users. Application Software basically includes Word Processors, Spreadsheets, etc. Types of Application software include General Purpose Software, Customized Software, etc.

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3	What is the purpose of compiler? Differentiate compiler and interpreter. (Mar-2023-OLD)[NLJIET]	3																
	• A compiler is a translator that converts the high-level language into the machine language. • High-level language is written by a developer and machine language can be understood by the processor. • Compiler is used to show errors to the programmer. • The main purpose of compiler is to change the code written in one language without changing the meaning of the program																	



	<table><tr><th>Compiler</th><th>Interpreter</th></tr><tr><td> • Compiler is a language processor which converts high level language to low level language and vise-versa. </td><td> • Interpreter is a language processor which converts high level language to low level language and vise-versa. </td></tr><tr><td> • It takes the entire program at a time. </td><td> • It takes a single line of code or instruction at a time. </td></tr><tr><td> • It generates intermediate object code. </td><td> • It does not produce any intermediate object code. </td></tr><tr><td> • The compilation is done before execution. </td><td> • Compilation and execution takes place simultaneously. </td></tr><tr><td> • Speed is Comparatively faster. </td><td> • Speed is slower. </td></tr><tr><td> • Memory requirement is more due to the creation of object code. </td><td> • It requires less memory as it does not create any intermediate object. </td></tr><tr><td> • Displays all errors after compilation, all at same time. </td><td> • Displays error of each line one by one. </td></tr><tr><td> • Error detection is difficult. </td><td> • Error detection is easier. </td></tr><tr><td> • It consumes less time. </td><td> • It consumes more time. </td></tr><tr><td> • Compiler is more efficient. </td><td> • Interpreter is less efficient. </td></tr><tr><td>Programming languages like; C, C++, C#, Scala, typescript uses a Compiler.</td><td>Programming languages like; PHP, Perl, Python, Ruby uses an Interpreter.</td></tr></table> <pre>graph LR; SC[/SOURCE CODE/] --> C[COMPILER]; C --> MC[MACHINE CODE]; MC --> O[/OUTPUT/];</pre><pre>graph LR; SC[/SOURCE CODE/] --> I[INTERPRETER]; I --> O[/OUTPUT/];</pre>	Compiler	Interpreter	• Compiler is a language processor which converts high level language to low level language and vise-versa.	• Interpreter is a language processor which converts high level language to low level language and vise-versa.	• It takes the entire program at a time.	• It takes a single line of code or instruction at a time.	• It generates intermediate object code.	• It does not produce any intermediate object code.	• The compilation is done before execution.	• Compilation and execution takes place simultaneously.	• Speed is Comparatively faster.	• Speed is slower.	• Memory requirement is more due to the creation of object code.	• It requires less memory as it does not create any intermediate object.	• Displays all errors after compilation, all at same time.	• Displays error of each line one by one.	• Error detection is difficult.	• Error detection is easier.	• It consumes less time.	• It consumes more time.	• Compiler is more efficient.	• Interpreter is less efficient.	Programming languages like; C, C++, C#, Scala, typescript uses a Compiler.	Programming languages like; PHP, Perl, Python, Ruby uses an Interpreter.	
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TOPIC:3 Concepts of Machine Level, Assembly Level and High Level Programming																										
SHORT QUESTIONS																										
1	Justify C is middle level language. [NLJIET]																									

	C is called middle-level language because it actually binds the gap between a machine level language and high-level languages. A user can use c language to do System Programming (for writing operating systems) as well as Application Programming (for generating menu driven customer billing systems). That's why it is called the middle-level language. C language is one of the most popular and oldest computer programming languages because it is a structured, machine-independent language and a high-level language. C language allow software developers to develop software or applications without worrying about the hardware feasibility, where they will be implemented.											
2	Define following terms: 1) Application Software 2) System Software.(Mar-2022-NEW) [NLJIET]	2										
	Software is a collection of instructions, procedures, and documentation that performs different tasks on a computer system. We can say also Computer Software is a programming code executed on a computer processor. The code can be machine-level code or code written for an operating system. Examples of software are MS- Word, Excel, PowerPoint, Google Chrome, Photoshop, MySQL, etc. Types of Computer Software • System Software • Application Software Types Of Software <pre>graph TD; A[Types Of Software] --> B[System Software]; A --> C[Application Software]; B --> D["Operating System E.g:- linux, windows etc."]; B --> E["Language Processor And Device Driver E.g:- Interpreter, compiler etc."]; C --> F["General purpose E.g:- MS-word, Photoshop etc."]; C --> G["Customised Software E.g:- Railways resevation system etc."];</pre> <table><thead><tr><th>System Software</th><th>Application Software</th></tr></thead><tbody><tr><td> • System software are mainly designed for managing system resources. </td><td> • Application software are designed to accomplish tasks for specific purposes. </td></tr><tr><td> • Programming of system software is complex. </td><td> • Programming of application software is comparatively easy. </td></tr><tr><td> • A computer cannot run without system software. </td><td> • A computer can easily run without application software. </td></tr><tr><td> • System software do not depend on application software. </td><td> • Application software depend on system software and cannot run without system software. </td></tr></tbody></table>	System Software	Application Software	• System software are mainly designed for managing system resources.	• Application software are designed to accomplish tasks for specific purposes.	• Programming of system software is complex.	• Programming of application software is comparatively easy.	• A computer cannot run without system software.	• A computer can easily run without application software.	• System software do not depend on application software.	• Application software depend on system software and cannot run without system software.	
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	DESCRIPTIVE QUESTIONS	
1	Define (1) Higher Level Language (2) Lower Level Language (3) Machine Level Language. (Jun-2019-OLD) [NLJIET]	3
	Higher Level Language: It is programmer friendly language. High-level languages are programming languages that are designed to allow humans to write computer programs and interact with a computer system without having to have specific knowledge of the processor or hardware that the program will run on. Lower Level Language: It is a type of language, which is machine- friendly. It needs assembler for translation. It is machine-dependent. Machine Level Language: The machine-level language is a language that consists of a set of instructions that are in the binary form 0 or 1. As we know that computers can understand only machine instructions, which are in binary digits, i.e., 0 and 1, so the instructions given to the computer can be only in binary codes.	



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